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Consortium Meeting 5

UNIMAS Digital Experience Hub, Universiti Malaysia Sarawak
19-20 October 2026

Advancements in Data Management: Technological Perspectives from Indian Institutions

Speaker: Maninder Singh

Institution: Thapar Institute of Engineering & Technology, Patiala

Project acronym: OPEN-ASIA
Project full title: OPEN-ASIA: Boosting engagement of HEIs in Open Science in India and Malaysia
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https://openasia-erasmus.org

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Today's roadmap

30 minutes · 6 themes · 2 interactive quiz breaks

01

Why data management matters now

3 min

02

How India's data landscape evolved

3 min

03

Thapar case: ecosystem, ERP & repository

7 min

Q1

Mentimeter quiz break 1

3 min

04

Technologies reshaping data management

4 min

05

Governance: DPDP Rules & Open Science

5 min

Q2

Mentimeter quiz break 2

3 min

06

Challenges & the way forward

2 min

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Why now? Data is national infrastructure

263.5 M

verified APAAR student IDs generated across India

as of 2 Jul 2026

2,899

higher education institutions registered on the Academic Bank of Credits

NAD-ABC portal

13,000+

journals opened to 6,300+ institutions via One Nation One Subscription

launched 1 Jan 2025

18

months of phased DPDP compliance after the Rules were notified on 14 Nov 2025

full compliance ≈ May 2027

Takeaway: every student record, credit, publication and dataset now sits inside a national, interoperable and regulated data fabric.

Sources: PIB factsheet on ABC & APAAR (Jul 2026); ONOS / INFLIBNET; MeitY DPDP Rules 2025 (PIB, Nov 2025).

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From paper registers to national data rails

Milestones that reshaped how Indian institutions manage data

2010

Shodhganga

National e-theses repository at INFLIBNET

2017

NAD

National Academic Depository for digital certificates

2021

ABC

Academic Bank of Credits under NEP 2020

2023

DPDP Act + APAAR

Data protection law and One Nation, One Student ID

2025

ONOS + DPDP Rules

National journal access (Jan) and privacy rules (Nov)

2026

NAD-ABC deadline

UGC: all HEIs upload credit data by 30 Jun 2026

Thapar in parallel: an in-house WebKiosk portal → PeopleSoft Campus Solutions ERP, a Moodle-based LMS, the DSpace-based TuDR repository and an e-Library portal (launched March 2023).

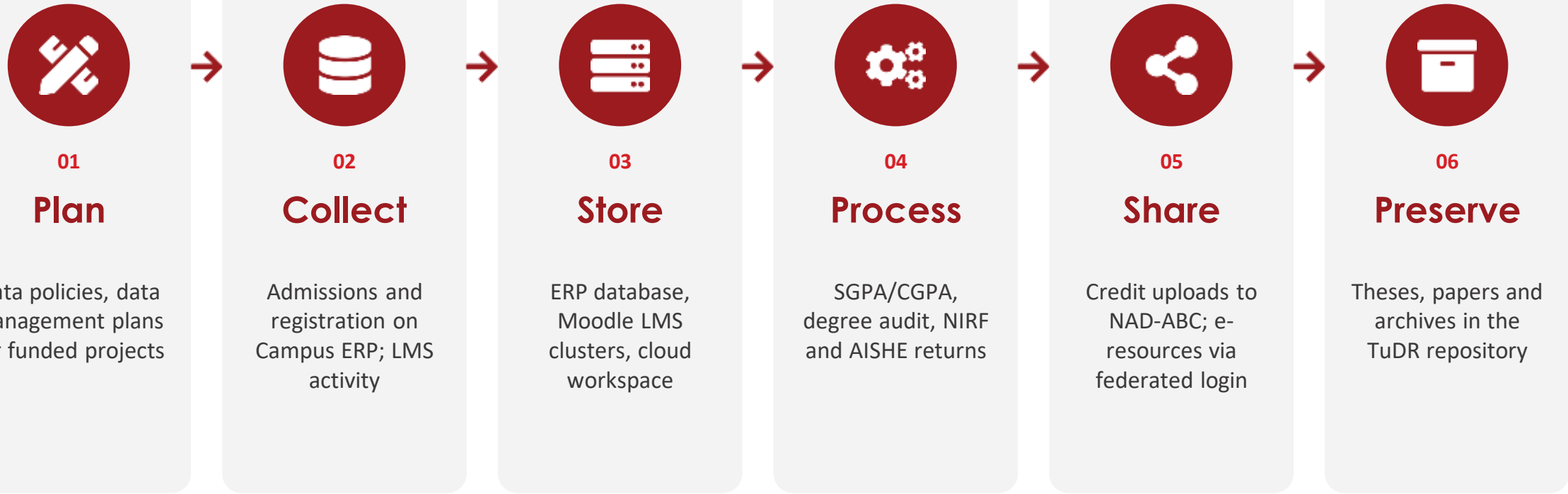
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A lifecycle lens on institutional data

Each stage mapped to where it happens at Thapar



Good data management means no stage is an afterthought, and preservation feeds planning for the next cycle.

Case study: Thapar Institute (TIET)

Why Thapar is a useful lens

- **Est. 1956, Patiala, Punjab**: a deemed-to-be university with seven decades of records
- **Hybrid landscape**: legacy and enterprise systems running side by side
- **Full research lifecycle**: LMS, ERP, library and an institutional repository

NIRF 2025 ranks

#29	#26	#44
Engineering	University	Overall

Source: NIRF 2025 rankings (Ministry of Education).

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Thapar's digital data ecosystem, layer by layer

Users — Students · Faculty · Researchers · Administrators

Identity & access — One @thapar.edu account (Google Workspace) · Shibboleth IdP for federated e-resource access

Campus ERP
PeopleSoft Campus Solutions: registration, grades, degree audit

WebKiosk
Legacy portal, run in parallel during migration

THAPAR-LMS
Moodle on four cluster servers

Central Library
Nava Nalanda Central Library + e-Library portal

TuDR
Institutional repository on DSpace 9.2

Support — eTicket: one helpdesk across IT (CITM), library, estate and ERP

National rails — ABC / APAAR · NAD–DigiLocker · NIRF & AISHE · ONOS · Shodhganga

Source: Thapar Digital Ecosystem guide (thapar.edu); tudr.thapar.edu; idp.thapar.edu.

Case in point: WebKiosk to Campus ERP

BEFORE

WebKiosk (in-house portal)

- Class schedules and attendance
- Exam dates and seating
- Results lookup
- Separate login from other systems



Parallel run

AFTER

PeopleSoft Campus Solutions

- Online course registration
- Exam schedules and grades
- SGPA/CGPA tracking
- Degree audit reports

De-risk the switch

Running both systems in parallel protects exams and results during migration.

Identity is the glue

One institutional account ties LMS, email and meetings together.

Next frontier: SSO

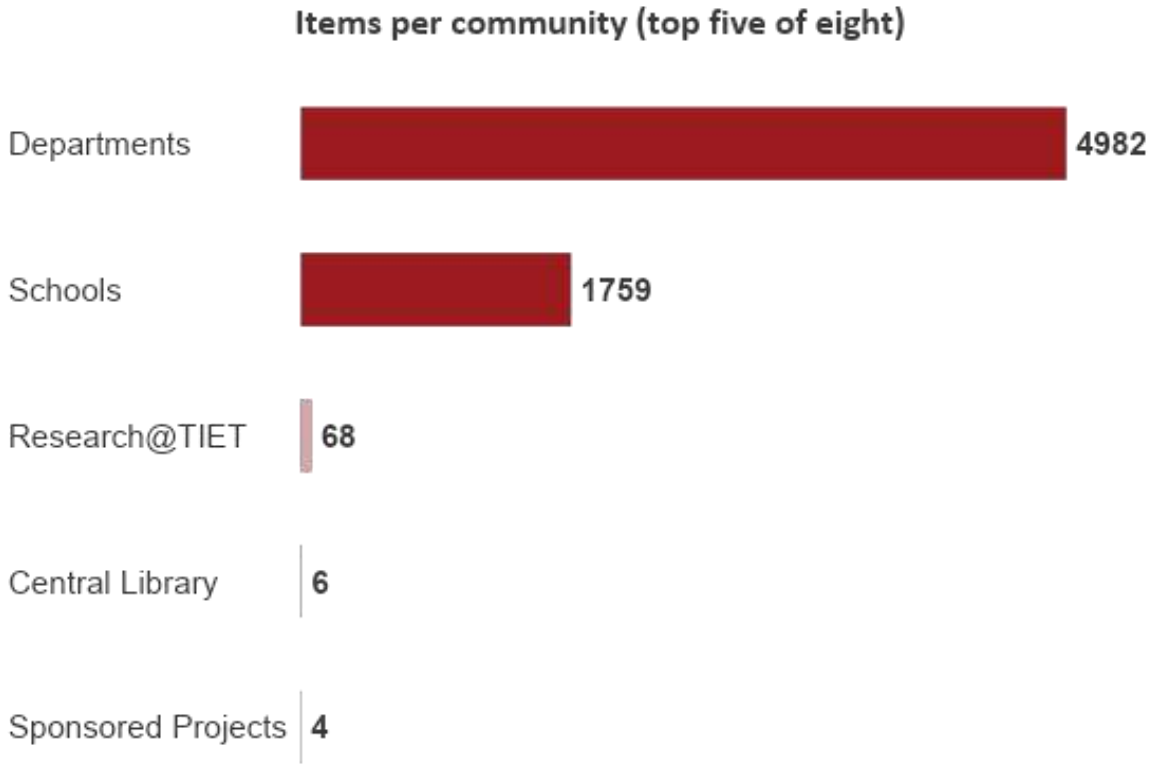
LMS and ERP still use separate credentials, a clear integration target.

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TuDR: Thapar's institutional memory



What the numbers tell us

6,800+

items across the top five communities: theses, papers and institutional records

Modern stack

DSpace 9.2 with persistent handles and OAI-PMH harvesting built in

Gap to close

only 4 items under Sponsored Projects, so research data deposit is the growth frontier

Source: tudr.thapar.edu home page, community item counts as displayed (Sep 2026).

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Quiz break 1: check your data IQ



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Technologies reshaping data management

Cloud & SaaS

India Institutional services move to managed platforms

Thapar One @thapar.edu Google Workspace identity for email, meetings and LMS access

Integrated ERP

India Samarth eGov ERP is offered to public HEIs

Thapar PeopleSoft Campus Solutions runs registration to degree audit

Learning analytics

India LMS click-streams inform teaching decisions

Thapar Moodle LMS logs attendance, quizzes and grades

AI & ML

India IndiaAI Mission and the AIKosh datasets platform

Thapar Opportunity: early-warning analytics on ERP and LMS data

Digital credentials

India NAD–DigiLocker certificates, APAAR, ABC credits

Thapar Credit data flows to the NAD-ABC portal

Open repositories

India Shodhganga and institutional repositories

Thapar TuDR on DSpace 9.2, harvestable via OAI-PMH

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Governance: DPDP Rules 2025 on campus

Key obligations

Notice & consent
Clear, itemised notices for students, staff and applicants

Children's data
Verifiable parental consent, with specific exemptions for education

Security safeguards
Encryption, access control, logs and monitoring

Breach reporting
Inform affected people and the Data Protection Board without delay

Rights & erasure
Access, correction and erasure requests handled within set timelines

14 Nov 2025 → May 2027

Rules notified, then an 18-month phased compliance window

Readiness checklist, using Thapar as the example

- [ ] Map personal data across ERP, LMS, library and email
- [ ] Role-based access and single sign-on
- [ ] Retention and deletion schedule per record type
- [ ] Breach playbook routed through CITM / eTicket
- [ ] Named grievance and data-protection contact

Source: MeitY, Digital Personal Data Protection Rules 2025 (PIB, 17 Nov 2025). Penalties under the DPDP Act go up to ₹2.5 billion (about US$30 million).

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Open Science: making research data FAIR

Where a DSpace repository like TuDR already helps, and the next step

F — Findable

TODAY

Persistent handles and rich metadata

NEXT STEP

Mint DOIs for datasets and link ORCID iDs

A — Accessible

TODAY

Open web access and standard protocols

NEXT STEP

Tiered access and embargoes for sensitive data

I — Interoperable

TODAY

Dublin Core metadata, OAI-PMH harvesting

NEXT STEP

Connect to a research information system (CRIS)

R — Reusable

TODAY

Clear provenance for theses and papers

NEXT STEP

Creative Commons licences and data management plans

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Quiz break 2: governance & Open Science



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Challenges Indian institutions face

Siloed systems

ERP, LMS, library and repository hold overlapping records with separate logins

Legacy migration

Parallel running doubles effort and risks data drift

Skills gap

Few trained data stewards; librarians are moving into data curation

Funding & licences

Enterprise ERP and cloud costs compete with academic priorities

Metadata quality

Inconsistent records weaken rankings, reporting and discoverability

Compliance load

DPDP, NIRF, AISHE and ABC reporting all run on the same data

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Way forward: a phased roadmap

0–6 months

Foundations

- Institution-wide data inventory
- DPDP gap analysis
- Single sign-on across ERP and LMS

6–18 months

Governance & Open Science

- Data governance council and stewards
- Research data collection in TuDR with DOIs
- DPDP compliance by May 2027

18–36 months

Intelligence

- Unified data platform for analytics
- AI-driven student success insights
- Open Science hub and national linkages

Sustainability: fund stewards as permanent roles, reuse national rails (ABC, ONOS, Shodhganga), and share practice with partner HEIs.

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Key takeaways

1

Data is infrastructure

APAAR, ABC, ONOS and DPDP make institutional data national, connected and regulated.

2

Thapar shows the path

Legacy to enterprise ERP, a Moodle LMS and a DSpace repository, all bound by one identity.

3

Governance unlocks value

FAIR research data and DPDP readiness come before AI.

Thank you

Questions & discussion

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